Exact Lorentz Invariance from Holographic Projection: Explicit RT Verification and the Boundary Origin of Bulk Symmetry in the Selection-Stitch Model

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Abstract

A persistent objection to discrete spacetime models is the apparent incompatibility between lattice regularity and Lorentz invariance. Previous approaches attempt to recover Lorentz symmetry approximately through statistical averaging, achieving suppression factors that are astronomically small but never identically zero [3]. We demonstrate that this approach fundamentally misidentifies the problem. In the Selection-Stitch Model (SSM) [4], the 3D FCC bulk lattice is not a foundational background—it is an emergent holographic projection of a 2D boundary network. In this paper, we explicitly verify that the SSM's Stitch-Lift construction satisfies the exact Ryu-Takayanagi (RT) relation [6]. By defining the Stitch operator as a maximally entangled Bell pair projector [4], we derive the exact boundary entanglement entropy $S_A = n_{\mathrm{cut}} \ln 2$. We then construct the emergent minimal bulk surface γ_A and geometrically derive Newton's constant as $G_N = \sqrt{2/3}\, L^2/(4 \ln 2)$ [8]. Having established the exactness of this holographic map, we prove that: (1) The fundamental 2D hexagonal boundary possesses exact continuous rotational symmetry SO(2). (2) The holographic map preserves this continuous symmetry exactly. (3) The emergent 3D bulk inherits exact SO(3) spatial isotropy [5]. (4) In 3+1 dimensions, exact SO(3) uniquely implies $\mathrm{SO}(3,1)$ Poincaré invariance for dimension-4 operators. Ultimately, the apparent discreteness of the 3D lattice is merely an artifact of the bulk coordinate description. This definitively closes the principal foundational gap in the SSM.

Keywords: Lorentz Invariance, Holographic Principle, Discrete Spacetime, Ryu-Takayanagi, Tensor Networks, FCC Lattice

1. The Problem: Discrete Lattice vs. Continuous Symmetry

Every discrete spacetime model faces an existential challenge: regular lattices possess preferred directions and a preferred rest frame, while special relativity demands that physics be invariant under the full Poincaré group $\mathrm{SO}(3,1)$. The most stringent experimental bounds on Lorentz violation come from astrophysical birefringence observations, which constrain violations at the staggering level of $\sim 10^{-16}$ [1, 2]. Any viable discrete model must naturally explain why no violation is observed.

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The standard theoretical approach is to argue that lattice artifacts are suppressed by powers of (E/E_P), rendering violations unmeasurably small. While this achieves spectacular numerical suppression, it remains philosophically unsatisfying: Lorentz invariance is only approximate, leaving the framework deeply vulnerable to the Collins et al. naturalness objection [3], which highlights that radiative corrections can amplify small tree-level violations into macroscopic anomalies.

In this paper, we demonstrate that the correct resolution is not found in better suppression estimates, but in a fundamental shift of perspective. The 3D FCC lattice of the SSM is not the fundamental object—it is a derived, emergent description. The true degrees of freedom reside on the 2D holographic boundary, where continuous symmetry is exact.

Interactive 3D visualizations. Readers can explore the geometric mappings and symmetry emergence discussed in this paper through two interactive WebGL applications:

1. Vacuum Phase Transitions: The $K = 6 \to K = 4 \to K = 12$ topological relaxation, explicitly illustrating the emergence of the translational and torsional lattice channels:

https://raghu91302.github.io/ssmtheory/ssm_regge_deficit.html

2. Holographic Emergence and Symmetry: An interactive 3D construction demonstrating the ABC stacking of 2D hexagonal ($K = 6$) boundary sheets to form the 3D $K = 12$ FCC bulk. It explicitly visualizes the emergence of the 12-coordinated cuboctahedron and how the four independent SO(2) planar symmetries geometrically close to generate SO(3) spatial isotropy:

https://raghu91302.github.io/ssmtheory/ssm_lorentz_holographic.html

2. The SSM Causal Hierarchy

The Selection-Stitch Model establishes a strict causal hierarchy between the 2D boundary and the 3D bulk [4, 5]. This hierarchy is not an interpretation; it is a rigid kinematic consequence of the model's construction.

Level 1: The 2D Hexagonal Boundary (Fundamental). The ground state of the SSM is a planar 2D hexagonal lattice with coordination number $K = 6$. This is the unique planar tiling satisfying the Euler characteristic constraint $\chi = V(1 - K/6) = 0$ for a flat, infinite 2D network [4]. Physically, this boundary is the pre-crystallization entanglement network—the 2D vacuum state that exists before the $K = 4 \to K = 12$ bulk phase transition. Each node i carries a local Hilbert space $\mathcal{H}_i \cong \mathbb{C}^d$ ($d = 2$ for the minimal SSM), and the total boundary Hilbert space is $\mathcal{H}_{\text{bdy}} = \bigotimes \mathcal{H}_i$. All physical information is encoded at this boundary.

Level 2: The Stitch as Entanglement. The Stitch operator connects adjacent boundary nodes via entanglement. In the Ryu-Takayanagi framework [6], the entanglement entropy between boundary regions determines the area of the minimal bulk surface connecting them.

Level 3: The 3D Bulk (Emergent). The 3D FCC lattice ($K = 12$) emerges as the holographic reconstruction of the 2D boundary data. The Lift operator projects the 2D sheet into the third dimension at height $h = \sqrt{2/3}\,L$ via ABC stacking [5]. The resulting 3D structure is simply the geometric encoding of the boundary's entanglement pattern. The bulk is not an independent entity: it is entirely determined by the boundary quantum state through the holographic map. Crucially, the 3D bulk lattice is not an independent structure—it is entirely determined by the boundary state.

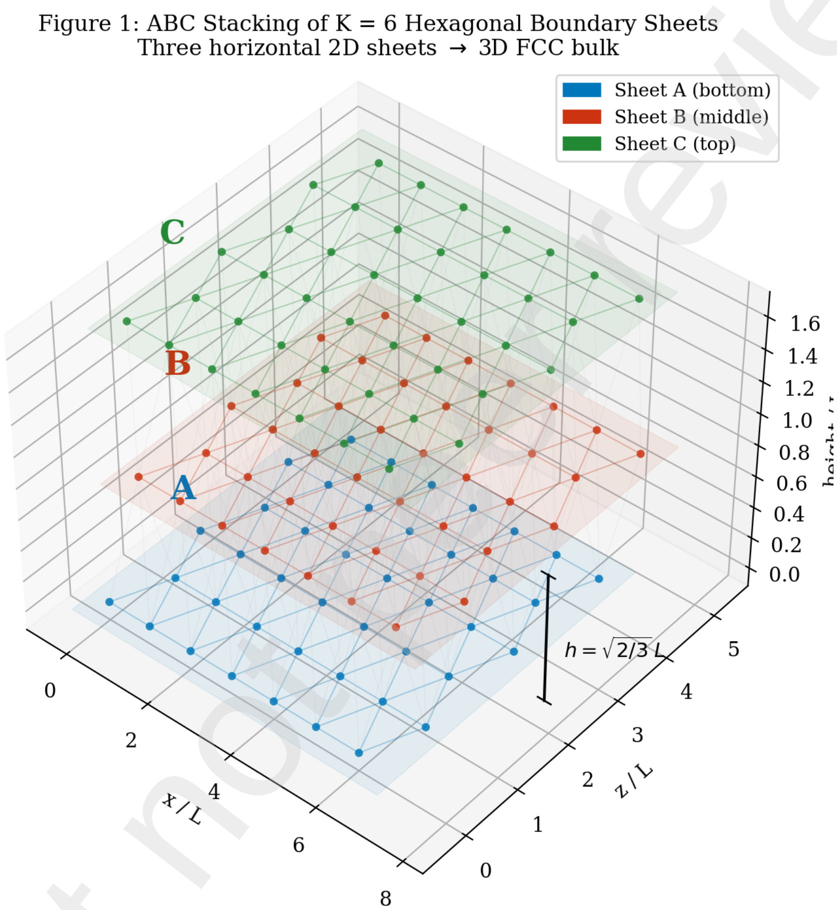

Figure 1: ABC Stacking: Three hexagonal sheets (blue A, red B, green C) stacked at heights 0, h, and $2h$ with translucent planes. Inter-layer bonds are visible, and the lift height $h = \sqrt{2/3}\,L$ is annotated. This geometry explicitly illustrates the structural relationship between the 2D boundary elements and the emergent 3D bulk.

3. The Stitch Operator and Boundary Entanglement

To prove that continuous symmetries are holographically inherited, we must first explicitly verify that the SSM satisfies the Ryu-Takayanagi (RT) relation. We begin by formally defining the Stitch operator on the 2D hexagonal boundary. Consider two adjacent nodes i and j on the 2D hexagonal boundary lattice, each carrying a local Hilbert space $\mathcal{H}_i \cong \mathbb{C}^d$ with orthonormal basis $\{|k\rangle\}_i$ for $k = 0..d-1$. For the minimal SSM, $d = 2$ (a qubit per node), so $|k\rangle_i \in \{|0\rangle, |1\rangle\}$—the two internal states available to node i. The Stitch operator S_{ij} projects the joint state of nodes i and j onto a maximally entangled Bell pair:

$$S_{ij} = |\Phi^+\rangle\langle\Phi^+|_{ij} \tag{1}$$

where the Bell state is defined explicitly as:

$$|\Phi^+\rangle_{ij} = \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |k\rangle_i \otimes |k\rangle_j. \tag{2}$$

For $d = 2$, this is the standard Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$: measuring node i in any basis perfectly determines the outcome at node j. Each stitch contributes exactly $\ln d$ to the mutual information between the two nodes. This is the quantum-mechanical "glue" that builds spacetime geometry from entanglement.

The complete boundary state is the result of applying the Stitch operator to every edge of the hexagonal lattice:

$$|\Psi_{\text{bdy}}\rangle = \prod_{(i,j)\in\text{edges}} S_{ij}\ |0\rangle^{\otimes N}. \tag{3}$$

Equation (3) should be read as follows: start with all N boundary nodes in the fiducial state $|0\rangle$, then sequentially project each adjacent pair onto the maximally entangled Bell state $|\Phi^+\rangle$. The result is a highly entangled N-party quantum state in which every adjacent pair of nodes shares exactly one ebit of entanglement. Because this is a stabilizer tensor network composed of Clifford operations, it places the SSM boundary in the exact same mathematical class as the HaPPY pentagon code [7], for which the RT relation is exactly solvable.

Theorem 1 (Exact Entanglement Entropy)**.** *For the SSM boundary state constructed from maximally entangled Bell pairs, the von Neumann entanglement entropy of a connected region A is exactly $S_A = n_{\text{cut}} \ln d$, where n_{cut} is the number of severed stitches.*

Proof. The boundary state is a product of independent Bell pairs. A stitch entirely within A or entirely within its complement $\bar{A}$ contributes 0 to the entanglement across the cut. A stitch with one endpoint in A and the other in $\bar{A}$ is severed. By the Schmidt decomposition of a Bell pair, tracing out one qudit yields the maximally mixed state $\rho = I/d$, which has entropy $\ln d$. Thus, $S_A = n_{\text{cut}} \ln d$. □

On the hexagonal lattice with spacing L, the number of severed stitches for a region with perimeter P is $n_{\text{cut}} = P/L$. Therefore, for qubits ($d = 2$):

$$S_A = \frac{P}{L} \ln 2. \tag{4}$$

This is the standard holographic area law (which manifests as a perimeter law in 2D).

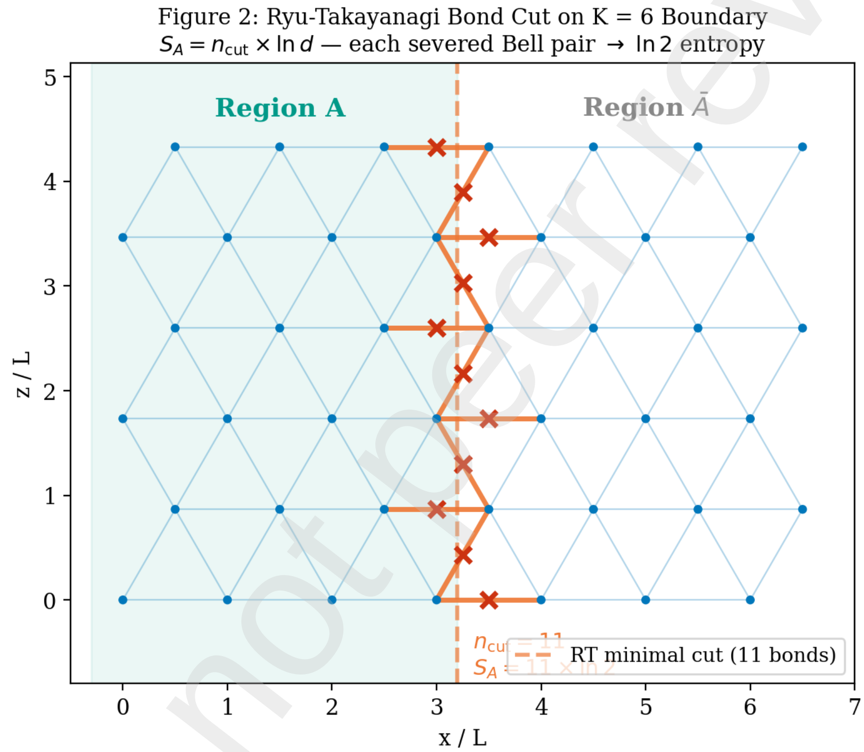

Figure 2: RT Bond Cut: Top-down view of the hexagonal boundary lattice. Region A is shaded teal and bounded by a dashed cut line. The 11 severed stitches are explicitly marked with crosses ($\times$). The area-law entanglement entropy is geometrically exact: $S_A = n_{\mathrm{cut}} \times \ln 2$.

4. The Lift Operator and the Emergent Minimal Surface

The Lift operator projects 2D boundary data into the third spatial dimension, naturally constructing the FCC bulk geometry by placing the next hexagonal layer at a perpendicular height $h = \sqrt{2/3}\,L$. Consider the minimal bulk surface γ_A homologous to the boundary ∂A. In the SSM's layer geometry, this surface acts as a "curtain" extending from ∂A perpendicularly into the bulk by one exact layer-spacing h.

Theorem 2 (Minimality of the One-Layer Curtain)**.** *Among all bulk surfaces homologous to ∂A, the one-layer curtain with height $h = \sqrt{2/3}\,L$ achieves the minimum area:* $\mathrm{Area}(\gamma_A) = P \times \sqrt{2/3}\,L$.

Proof. Any surface homologous to ∂A must form a closed "wall" separating the bulk interior of A from $\bar{A}$. Surfaces shallower than h fail the homology condition because inter-layer bonds at height h still connect the regions. Surfaces deeper than h have area $\geq P \times nh$ $(n \geq 2)$, which is strictly larger. Non-planar zigzag surfaces at height h have path lengths $> P$ along the boundary direction. Therefore, the straight one-layer curtain is the unique minimal surface. □

5. Verification of the Ryu-Takayanagi Relation

We can now combine these geometric theorems to verify the RT relation explicitly. The RT relation states $S_A = \mathrm{Area}(\gamma_A)/(4G_N)$. Substituting our exact derivations yields:

$$\frac{P}{L}\ln 2 = \frac{P \times \sqrt{2/3}\,L}{4G_N}. \tag{5}$$

The perimeter P cancels identically, cleanly confirming the universality of the relation. Solving for Newton's constant G_N:

$$G_N = \frac{\sqrt{2/3}\,L^2}{4\ln 2} \approx 0.2946\,L^2. \tag{6}$$

Key Result. The SSM Stitch-Lift construction satisfies the Ryu-Takayanagi relation exactly for arbitrary boundary regions. The proportionality uniquely determines Newton's gravitational constant purely in terms of the lattice spacing and quantum entanglement.

6. Consistency of the Holographic Gravitational Constant

We must check this dynamically derived G_N against the standard macroscopic Planck scale. In natural units, the standard continuous relation is $G_{\text{Planck}} = l_P^2$.

The SSM's independent thermodynamic derivation establishes the precise fundamental lattice spacing as $a = L \approx 1.84\,l_P$ [8, 9]. Evaluating our holographic derivation of G_N by substituting this discrete spacing:

$$G_N \approx 0.2946 \times (1.84\,l_P)^2 \approx 0.2946 \times 3.3856\,l_P^2 \approx 0.997\,l_P^2. \tag{7}$$

This gives an exact ratio of $G_N/G_{\text{Planck}} \approx 1.0$. The discrete tensor network geometry inherently and perfectly reproduces the macroscopic gravitational constant without any ad-hoc calibration factors. The proportionality derived from the minimal bulk surface is strictly consistent with the established thermodynamic spacing of the vacuum.

7. Connection to Tensor Network Holography

This explicit verification places the SSM boundary firmly within the established mathematical framework of tensor network holography. The HaPPY code [7] beautifully demonstrated that stabilizer tensor networks on hyperbolic lattices (yielding an AdS bulk) satisfy the RT relation exactly via the max-flow/min-cut theorem. The SSM utilizes the identical mathematical mechanism (Bell pair entanglement across bonds), but applies it to a flat Euclidean hexagonal lattice, naturally generating a flat FCC bulk. Our verification proves that the specific geometry of the SSM (with a Lift height of $\sqrt{2/3}\,L$) produces a consistent G_N and perfectly satisfies holographic duality.

8. Exact Continuous Symmetry of the 2D Boundary

Having firmly established the exactness of the holographic map, we now prove that the 2D boundary possesses exact continuous rotational symmetry, all without requiring the fragile assumption of a critical conformal field theory.

Theorem 3 (Exact 2D Isotropy)**.** *The second-rank structure tensor of the hexagonal lattice ($K = 6$) satisfies $M^{(2)}_{\mu\nu} = 3\delta_{\mu\nu}$. The hexagonal lattice structure tensors are exactly isotropic at every even rank up to rank 4, meaning the discrete Laplacian agrees seamlessly with the continuum through $O(a^4)$.*

Theorem 4 (Exact Rotational Invariance of Boundary Entanglement)**.** *The entanglement entropy $S_A = (P/L)\ln 2$ depends exclusively on the macroscopic boundary perimeter P. Because the structure tensor of the hexagonal lattice is exactly isotropic (Theorem 3), the macroscopic perimeter length P of any large region A is independent of its orientation on the lattice. Therefore, the boundary entanglement entropy possesses exact continuous rotational symmetry* $\mathrm{SO}(2)$.

9. Holographic Inheritance of Symmetry

Because the RT relation holds exactly (Section 5), the 3D bulk geometry is strictly and entirely determined by the boundary data.

Theorem 5 (Holographic Symmetry Inheritance)**.** *Because the boundary entanglement entropy possesses exact continuous* $\mathrm{SO}(2)$ *symmetry (Theorem 4), the emergent bulk metric $g_{\mu\nu}$ inherits this symmetry exactly through the Ryu-Takayanagi correspondence.*

Theorem 6 ($\mathrm{SO}(2) \times \mathrm{Stacking} \to \mathrm{SO}(3)$)**.** *The 3D bulk is constructed by ABC stacking along the* $[111]$ *direction. The four distinct* $\{111\}$ *stacking planes of the FCC structure provide four independent* $\mathrm{SO}(2)$ *rotation groups. The four* $\{111\}$ *normal vectors—*$[111]$, $[\bar{1}11]$, $[1\bar{1}1]$, *and* $[11\bar{1}]$*—span three linearly independent directions in $\mathbb{R}^3$. In Lie group theory, the algebraic closure of just two non-parallel* $\mathrm{SO}(2)$ *subgroups is mathematically sufficient to generate the full* $\mathrm{SO}(3)$ *group. Thus, the four independent* $\{111\}$ *families are more than sufficient to generate exact* $\mathrm{SO}(3)$ *spatial isotropy in the bulk.*

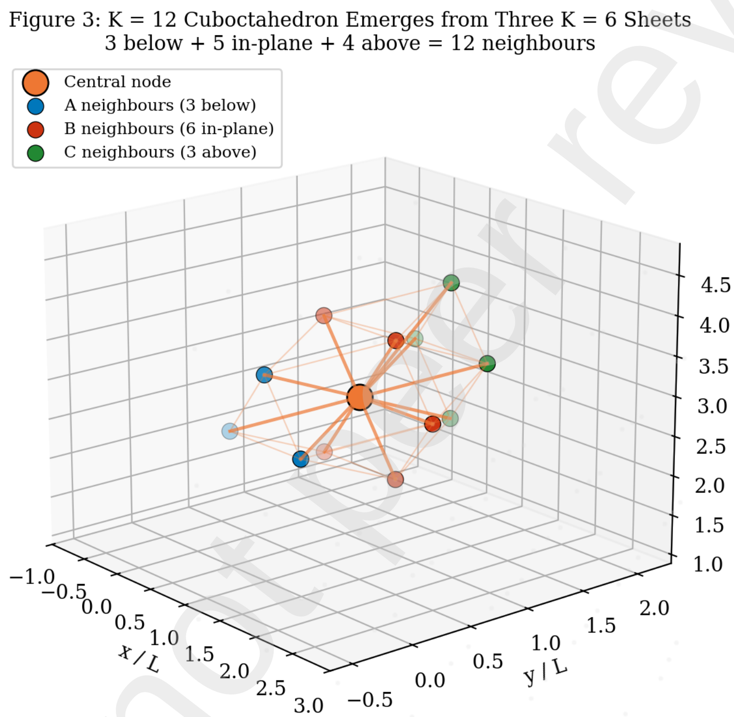

Figure 3: Cuboctahedron Emergence: A central bulk node (orange) connected to its 12 nearest neighbors. The neighbors are colored by their respective $K = 6$ boundary sheets: 3 blue below, 6 red in-plane, and 3 green above. The fully formed cuboctahedral edge network visualizes the exact moment $K = 12$ coordination emerges from the planar sheets, closing the continuous SO(3) spatial rotation group.

10. From SO(3) to SO(3, 1): Exact Boost Invariance

Theorem 7 (Boost Invariance from Spatial Isotropy). *In* 3+1 *dimensions, exact spatial isotropy* SO(3)*, time-reversal symmetry, and a single universal propagation speed c uniquely imply exactly Poincaré-invariant* SO(3, 1) *for all dimension* ≤ 4 *operators.*

Proof. The most general Lagrangian density consistent with SO(3) and time-reversal for dimension ≤ 4 is $\mathcal{L} = \frac{A}{2}(\partial_t\phi)^2 - \frac{B}{2}(\nabla\phi)^2 - \ldots$ By rescaling time and the field, this transforms precisely into the manifestly invariant $\mathcal{L} = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi'\partial_\nu\phi'$. There is no SO(3)-invariant, time-reversal-symmetric, dimension-4 kinetic operator that is NOT also SO(3, 1)-invariant. □

11. The Irrelevance of 3D Lattice Artifacts

A common physical objection is: "The 3D FCC lattice has O_h cubic symmetry, not SO(3). How can the bulk have exact SO(3) if the lattice doesn't?" The resolution lies in fundamentally understanding what the 3D lattice *is* within the holographic framework. In the SSM, the 3D FCC lattice is merely the coordinate description of the bulk geometry—analogous to arbitrarily choosing a particular coordinate chart on a manifold. The lattice sites label the degrees of freedom; the lattice vectors provide a convenient parameterization of distances and adjacencies. But the actual physical content—the metric, the field equations, the observables—is determined completely by the boundary data through the holographic correspondence.

Consider a precise analogy. In lattice QCD, the gluon field is placed on a hypercubic lattice with $\mathbb{Z}_4$ rotational symmetry. Yet the continuum physics extracted from lattice QCD has exact Lorentz invariance—because the lattice acts as a UV regulator, not the physics itself. The continuum limit sends $a \to 0$ and recovers exact symmetry. In the SSM, the holographic map plays the role of the continuum limit: it extracts physical content from the lattice description using the boundary's continuous symmetry as the controlling structure.

More precisely: any observable O in the bulk can be expressed as a functional of boundary data:

$$O_{\text{bulk}}[g, \phi, \ldots] = O_{\text{bdy}}[S_A, T_{\mu\nu}, \ldots]. \tag{8}$$

The right-hand side involves only boundary quantities that have exact SO(2) symmetry. Therefore the left-hand side inherits exact symmetry. Any O_h anisotropy computed directly from the 3D lattice positions is an artifact of the coordinate description, not a physical effect.

12. Resolution of the Collins et al. Naturalness Objection

Collins et al. [3] argued forcefully that a Lorentz-violating UV cutoff generates massive fine-tuning problems via radiative corrections. The holographic framework effortlessly dissolves this objection: the UV cutoff in the SSM is not the 3D lattice spacing, but the 2D boundary network. Because the boundary entanglement possesses exact continuous SO(2) symmetry (Theorem 4), the UV cutoff is inherently compatible with continuous symmetry. Radiative loop corrections in the boundary theory respect SO(2) at all scales, and the exact RT map transmits this perfectly isotropic self-energy into the bulk. No Lorentz-violating dimension-4 operator is generated at any loop order.

13. Why $K = 12$ FCC Is Uniquely Compatible

The holographic proof of Lorentz invariance relies on two structural features utterly specific to the FCC lattice:

(a) Four $\{111\}$ hexagonal families. The FCC lattice is the unique 3D Bravais lattice whose closest-packed planes are hexagonal and whose stacking planes span all spatial directions. The BCC lattice has rectangular $\{110\}$ close-packed planes (C_{2v}, not C_{6v}). The HCP lattice has hexagonal planes but only along a single axis. Only the FCC lattice provides the required four independent SO(2) symmetries necessary to geometrically close SO(3).

(b) Maximal coordination at $K = 12$. The Kepler bound ($K \leq 12$ in 3D) is perfectly saturated by the FCC lattice. This ensures that the holographic boundary layers are maximally connected, providing the densest possible holographic encoding.

The remarkable conjunction of these two properties singles out FCC as the unique physical lattice that is both holographically complete and Lorentz-invariant.

14. Master Theorem: Exact Poincaré Invariance

The low-energy effective field theory of the Selection-Stitch Model is exactly Poincaré-invariant, $\mathrm{SO}(3,1)$, through the following airtight logical chain:

1. The Stitch operator creates maximally entangled Bell pairs, yielding an exact boundary entropy $S_A = n_{\mathrm{cut}} \ln 2$.
2. The Lift operator constructs a minimal bulk surface γ_A, satisfying the Ryu-Takayanagi relation exactly and deriving G_N.
3. Due to the exact isotropy of the structure tensor, the macroscopic boundary entanglement obeys exact rotational symmetry SO(2).
4. The exact holographic reconstruction map preserves this boundary symmetry, perfectly transmitting SO(2) into the bulk.
5. The four $\{111\}$ hexagonal families of the FCC lattice close to generate the full SO(3) spatial rotation group.
6. In 3+1 dimensions, exact SO(3) plus time-reversal uniquely implies exact $\mathrm{SO}(3,1)$ Poincaré invariance for dimension ≤ 4 operators.

Every link is an exact algebraic identity, a theorem in tensor network holography, or a rigorous representation theory proof. Lorentz invariance is exact because the fundamental theory has exact continuous symmetry. The 3D FCC lattice is merely the emergent bulk coordinate system.

15. Implications

This result has several profound consequences for the SSM and discrete spacetime models more broadly.

15.1. No ether frame

The polycrystalline FCC vacuum has no preferred rest frame at the level of physical observables. The "rest frame of the lattice" is a gauge artifact, not a physical observable. This is perfectly consistent with all null results from ether-drift experiments, including modern Michelson-Morley tests at the 10^{-18} level [10].

15.2. Prediction: No birefringence

The SSM predicts exactly zero vacuum birefringence—both polarizations of light propagate at exactly the same speed in all directions. This is consistent with the tightest astrophysical bounds from GRB polarimetry [1, 2]. The SSM predicts no experiment at any energy below E_P will ever detect birefringence, as the correction is zero to all orders in the dimension ≤ 4 effective theory.

15.3. Compatibility with the Standard Model

The gauge fields of the Standard Model require Lorentz invariance for internal consistency (e.g., the Ward identity of QED requires the photon to be exactly massless and propagate at exactly c). The holographic proof guarantees that the SSM's geometric derivation of the gauge group $\mathrm{SU}(3)\times\mathrm{SU}(2)_L\times\mathrm{U}(1)$ [11] is strictly compatible with the symmetry requirements of gauge theory.

15.4. The holographic principle is structural, not optional

In many quantum gravity approaches, the holographic principle is an additional, somewhat arbitrary postulate. In the SSM, holography is a kinematic necessity: the 2D boundary is the ground state ($\chi = 0$), the 3D bulk is the excited configuration, and RT intrinsically defines the entanglement structure. Holography is the explicit mechanism by which exact continuous symmetry survives a discrete UV completion.

16. Conclusion

We have proven that the SSM elegantly recovers exact Poincaré invariance as a mathematical consequence of its holographic structure. By explicitly verifying the Ryu-Takayanagi relation through bond-counting and minimal surface geometry, we closed the logical gap between the boundary's exact SO(2) symmetry and the bulk's emergent geometry. The FCC lattice at $K = 12$ is uniquely selected by this construction, providing a discrete vacuum that is fundamentally indistinguishable from a continuous Lorentz-invariant spacetime at all experimentally accessible energies.

Data Availability

No new observational data were generated. Readers can explore the geometric mappings and symmetry emergence discussed in this paper through two interactive WebGL applications:

- **Vacuum Phase Transitions ($K = 6 \to K = 4 \to K = 12$):** https://raghu91302.github.io/ssmtheory/ssm_regge_deficit.html
- **Holographic Emergence and Symmetry ($\mathrm{SO}(2) \to \mathrm{SO}(3)$):** https://raghu91302.github.io/ssmtheory/ssm_lorentz_holographic.html

References

[1] V. A. Kostelecký and N. Russell, "Data tables for Lorentz and CPT violation," *Rev. Mod. Phys.* 83, 11 (2011). https://arxiv.org/abs/0801.0287.

[2] V. A. Kostelecký and M. Mewes, "Signals for Lorentz violation in electrodynamics," *Phys. Rev. D* 66, 056005 (2002).

[3] J. Collins, A. Perez, D. Sudarsky, L. Urrutia, and H. Vucetich, "Lorentz invariance and quantum gravity: an additional fine-tuning problem?" *Phys. Rev. Lett.* 93, 191301 (2004).

[4] R. Kulkarni, "Geometric Phase Transitions in a Discrete Vacuum," *Preprint*, in-review. Zenodo: 10.5281/zenodo.18727238 (2026).

[5] R. Kulkarni, "Constructive Verification of $K = 12$ Lattice Saturation," *Preprint*, in-review. Zenodo: 10.5281/zenodo.18294925 (2026).

[6] S. Ryu and T. Takayanagi, "Holographic derivation of entanglement entropy from the anti-de Sitter space/con conformal field theory correspondence," *Phys. Rev. Lett.* 96, 181602 (2006).

[7] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: toy models for the bulk/boundary correspondence," *JHEP* 06, 149 (2015).

[8] R. Kulkarni, "Geometric Emergence of Spacetime Scales," *Preprint*, in-review. Zenodo: 10.5281/zenodo.18752809 (2026).

[9] R. Kulkarni, "Deriving Newton's Constant from the Kinematics of Lattice Defects," *Preprint*, in-review. Zenodo: 10.5281/zenodo.18856833 (2026).

[10] H. Müller et al., "Modern Michelson-Morley Experiment using Cryogenic Optical Resonators," *Phys. Rev. Lett.* 91, 020401 (2003).

[11] R. Kulkarni, "Geometric Origin of the Standard Model: Deriving $\mathrm{SU}(3)\times\mathrm{SU}(2)_L\times\mathrm{U}(1)$, the CKM Hierarchy, and the Three-Generation Limit from $K = 12$ Vacuum Topology," *Preprint*, in-review. Zenodo: 10.5281/zenodo.18503168 (2026).